

★ WEEK 16 · PART 1 · PRACTICE

Choosing a Model, Worked by Hand

Grade Forecasts Out-of-Sample; Match the Shape to the Decision

Naive vs AR(1) on a scoreboard - and a point vs a distribution

Part 1's two big ideas, with the arithmetic shown. We grade two forecasts on held-out data and let the numbers pick a winner, then compute a tail risk a point forecast can never reveal.

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PART 1 · APPLICATIONS

Two Cases, Computed by Hand

Case A: forecast a return two ways, grade both on four held-out days, and let RMSE decide. Case B: turn a mean and a volatility into a tail probability and a worst-case loss. Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

Three families, three forecast shapes

A point, a regime, or a full distribution. Which one you need depends on the decision in front of you.

2

There is no universal best model

The competitions taught the field to test, not argue. We will grade two forecasts on held-out data and let the numbers decide.

3

Match the shape to the decision

A point forecast answers 'what return?'; only a distribution answers 'how much can I lose?'. We compute both by hand.

Everything here is a calculator exercise: four held-out days, one mean, one volatility.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the key numbers by hand, step by step.
- Then read off the decision it supports.

What you should watch

- How a held-out test crowns a winner.
- How a point forecast stays silent on risk.
- How the distribution answers the risk question.

Grade the Forecasts Out-of-Sample

Let the held-out data crown the winner

A desk needs a next-day return forecast for a signal. Two candidates: the naive forecast (tomorrow equals today) and an AR(1) model. Which is better? We do not argue - we grade both on four days neither saw, and let the errors decide.

The two candidates

- Naive: repeat the last observed return.
- AR(1): scale the last return by a coefficient.
- The naive forecast is the bar to beat.

What we'll do by hand

- Forecast four held-out days each way.
- Compute the errors and their squares.
- Compare RMSE and MAE, and pick a winner.

Naive vs AR(1)

The rules we will grade

$$\text{naive: } \hat{r}_t = r_{t-1} \qquad \text{AR(1): } \hat{r}_t = \phi r_{t-1}, \quad \phi = 0.5$$

The naive forecast

- Tomorrow's return equals today's.
- No parameters, no fitting.
- Surprisingly hard to beat.

The AR(1) forecast

- Scale the last return by $\phi = 0.5$.
- ϕ was estimated on the training data.
- More sophisticated - but is it better?

The Held-Out Test: Four Days

Forecast each day two ways

Day	$r(t-1)$	actual $r(t)$	naive fc	AR(1) fc
1	+1.0	+0.4	1.0	0.5
2	+0.4	-0.6	0.4	0.2
3	-0.6	+0.8	-0.6	-0.3
4	+0.8	-0.2	0.8	0.4

Goal: score each forecast by how far it lands from the actual. $AR(1) \text{ fc} = 0.5 \times r(t-1)$; $\text{naive fc} = r(t-1)$.

The Errors

Actual minus forecast, then squared

naive errors: $-0.6, -1.0, +1.4, -1.0 \Rightarrow \text{squared sum} = 4.32$
AR(1) errors: $-0.1, -0.8, +1.1, -0.6 \Rightarrow \text{squared sum} = 2.22$

The AR(1) errors are smaller day by day, so its squared-error sum (2.22) is far below the naive's (4.32).

RMSE, MAE, and the Winner

Turn errors into a score

$$\text{RMSE}_{\text{naive}} = \sqrt{4.32/4} = 1.04 \quad \text{RMSE}_{\text{AR}} = \sqrt{2.22/4} = \mathbf{0.74}$$

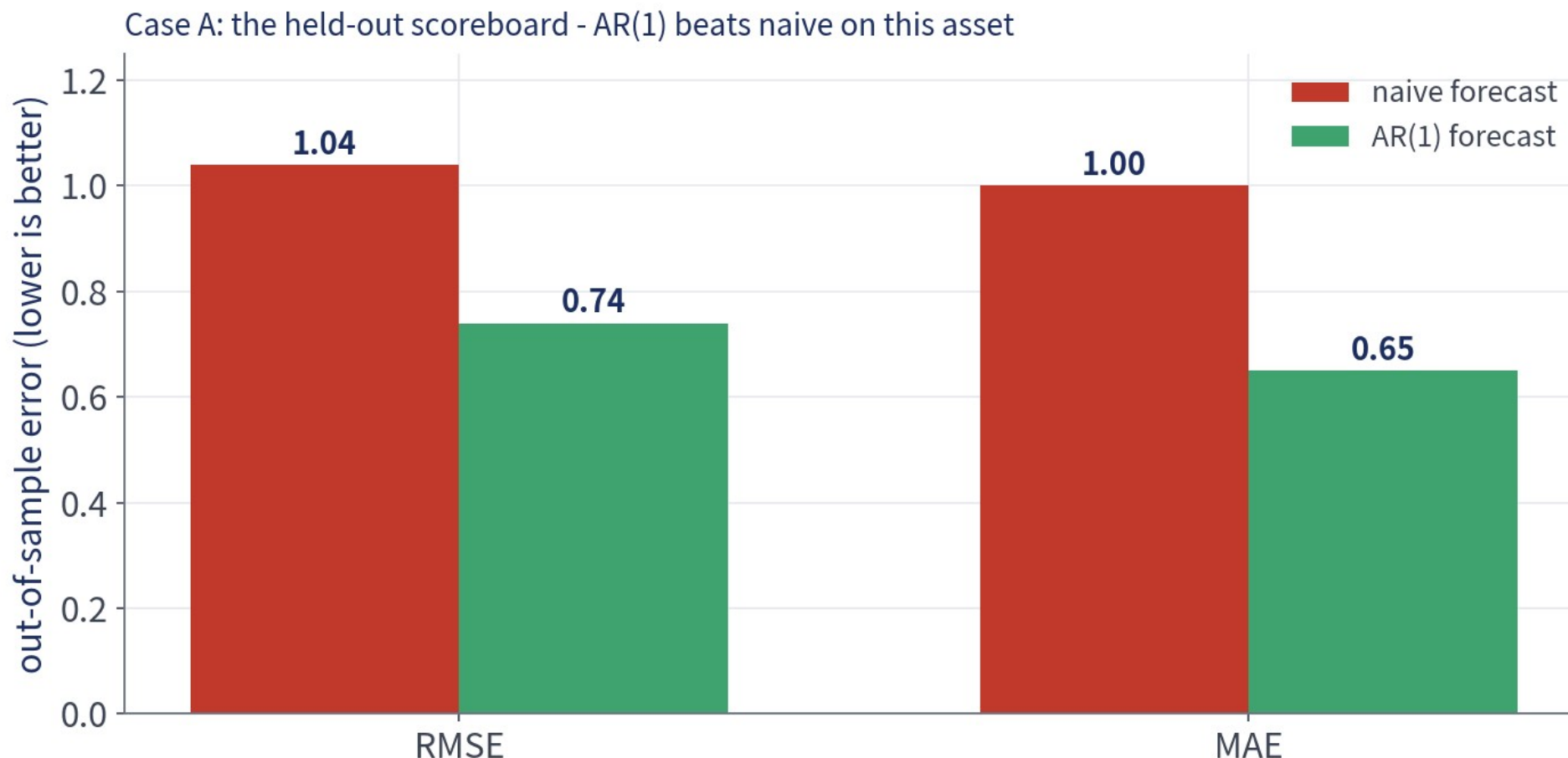
$$\text{MAE}_{\text{naive}} = 4.0/4 = 1.00 \quad \text{MAE}_{\text{AR}} = 2.6/4 = \mathbf{0.65}$$

$\Rightarrow$ AR(1) wins the held-out test - on this asset

On this asset AR(1) wins - but the point is the METHOD: on another asset the naive might win, which is exactly why you test.

The Held-Out Scoreboard

The same numbers as a picture: on both RMSE and MAE, the AR(1) forecast sits below the naive one for this asset. Lower is better - and this held-out comparison, not in-sample fit, is what settles the choice.



Why the Naive Forecast Is the Benchmark

Beat it, or don't bother

The naive forecast

- Costs nothing and needs no fitting.
- For many financial series it is very strong.
- It is the honest bar every model must clear.

The discipline

- Always report your model's error next to naive.
- If you can't beat naive, prefer the simpler one.
- Sophistication is only worth it if it scores better.

What the Test Decides

Don't trust a fit you can't defend

The wrong way

- Crown the model with the best in-sample fit.
- Assume the fancier model must forecast better.
- Never check it on data it hasn't seen.

The right way

- Fit on the past, grade on the held-out future.
- Compare against the naive benchmark.
- Let the out-of-sample score pick the winner.

When a Point Isn't Enough

A risk decision needs the whole distribution

Now a different decision: how much could this position lose next week? A point forecast of +0.5% sounds reassuring - but it says nothing about the downside. For a risk decision you need the distribution, and we compute its tail by hand.

The decision

- Set risk capital for a bad week.
- You need the tail, not the average.
- A single number cannot answer this.

What we'll do by hand

- Take the mean and the volatility.
- Compute the 5% worst-case return.
- Compute the probability of a big loss.

A Point vs a Distribution

Same forecast, two very different answers

The point forecast says

- 'Next week: about +0.5%.'
- Comforting, and completely silent on risk.
- It hides the whole left tail.

The distribution says

- '+0.5% on average, but with this spread.'
- It names the worst case and its odds.
- That is what a risk limit is built on.

The Forecast: Mean and Vol

Two numbers to build a distribution

Forecast input	Value
mean return (μ)	+0.5%
volatility (σ)	2.0%

Assume next-week returns are roughly normal with this mean and volatility. Goal: find the 5% worst case and the odds of a big loss.

The 5% Worst Case

One line from mean and vol

$$\begin{aligned} \text{forecast: mean } \mu &= +0.5\%, \text{ vol } \sigma = 2.0\% \\ 5\% \text{ worst case} &= \mu - 1.645 \sigma = 0.5 - 1.645(2.0) = -\mathbf{2.79\%} \end{aligned}$$

There is a 5% chance next week's return is worse than -2.79%. The point forecast of +0.5% never hinted at this.

The Odds of a Big Loss

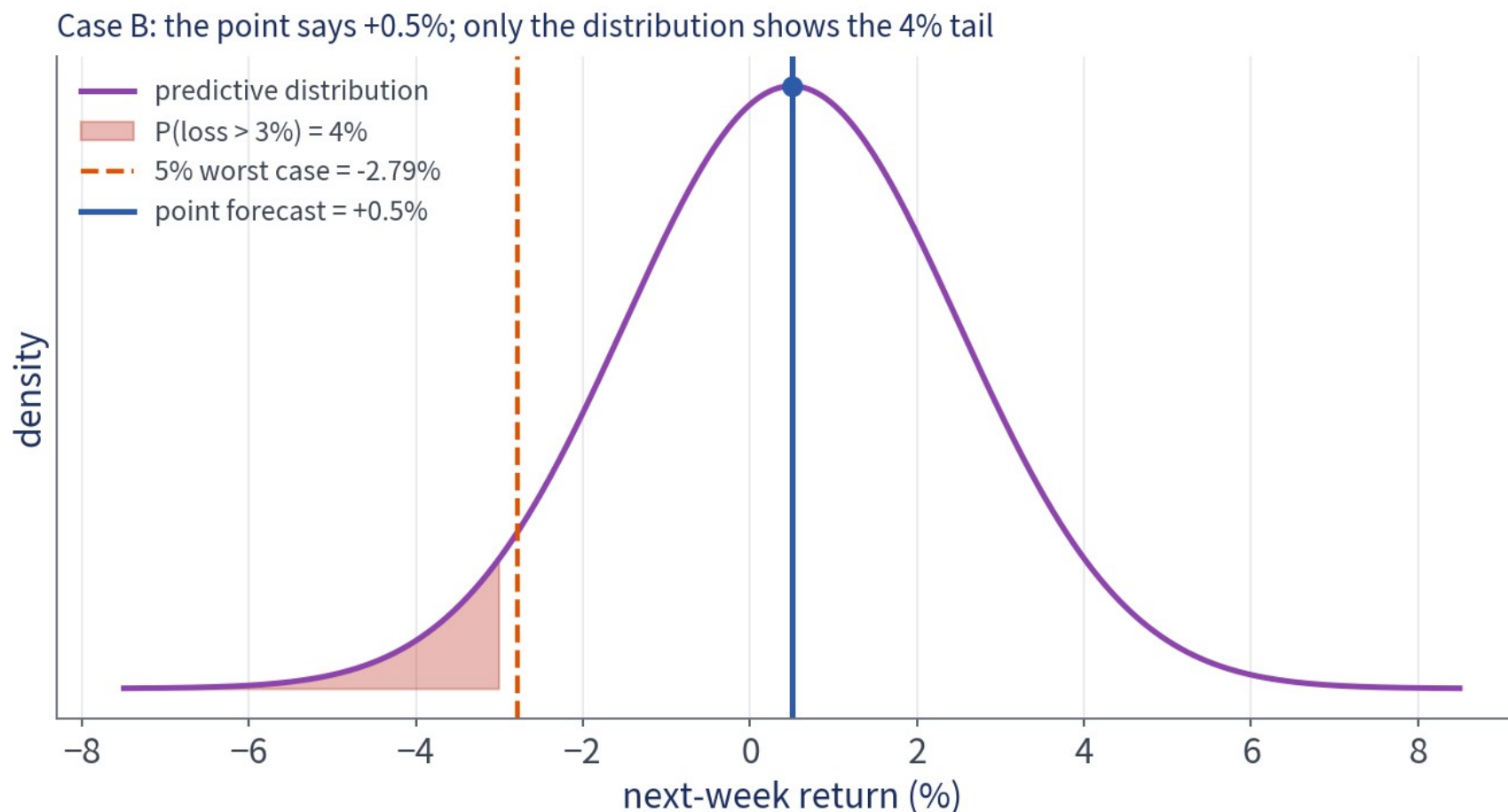
A z-score and a table lookup

$$\begin{aligned} P(r < -3\%) &= P\left(Z < \frac{-3-0.5}{2.0}\right) = P(Z < -1.75) \\ &= \Phi(-1.75) = 0.040 = 4\% \end{aligned}$$

Standardize the threshold to a z-score, then read the normal table: about a 4% chance of losing more than 3% in the week.

The Point Hides the Tail

The point forecast is one dot at +0.5%. The predictive distribution shows the whole picture: a 5% worst case near -2.79% and a shaded 4% chance of losing more than 3%. Only the distribution answers the risk question



Match the Shape to the Decision

Which family, for which question

If the decision is...

- 'What return do I book?' - a point is fine.
- 'Am I in a safe regime?' - you need the regime.
- 'How much can I lose?' - you need the distribution.

...then the family is

- Classical: the point forecast from Case A.
- State-space: the regime probability.
- Bayesian: the predictive tail from Case B.

What the Distribution Buys You

Risk you can size against

With only the point

- You see +0.5% and feel safe.
- You cannot set a risk limit from it.
- The tail catches you by surprise.

With the distribution

- You quote the 5% worst case: -2.79%.
- You size risk capital to the tail.
- You are ready for the bad week.

The Common Pattern

Two calculations, one discipline

Both cases, computed by hand, enforce the same discipline. Don't assume a model is best - grade it on held-out data. Don't accept a point when the decision needs the tail - compute the distribution. The right answer always starts from the decision.

The trap

- Trusting in-sample fit or a lone number.
- Assuming complex beats simple.
- Answering a risk question with an average.

The fix

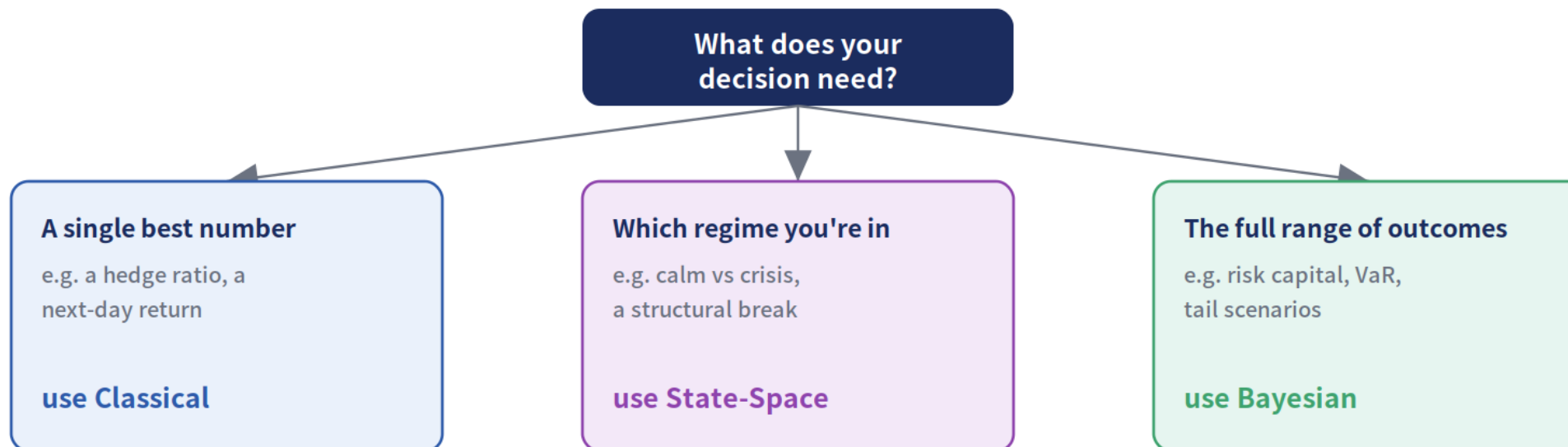
- Grade every model out-of-sample.
- Beat the naive benchmark or simplify.
- Match the forecast shape to the decision.

Let the Decision Choose the Model

Case A gave a point for a signal; Case B gave a distribution for risk. The map is the same rule: a single number points to Classical, the regime to State-Space, the full range to Bayesian - and out-of-sample testing breaks ties

Let the Decision Choose the Model

There is no universal best model. Ask what your decision actually needs, then pick the family.



And often: run all three, then let out-of-sample accuracy break the tie.

The families are complements, not rivals - the capstone compares them fairly on one asset.

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both worked examples in code, then extend them to a real hold-out.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - grade forecasts:

- * take the 4-day table; form naive and AR(1) forecasts; print each error and squared error;
- * compute RMSE and MAE for both; name the winner;
- * then repeat on a real asset with a train/test split and compare to the naive benchmark.

CASE B - point vs distribution:

- * from $\mu = 0.5\%$, $\sigma = 2\%$, print the 5% worst-case return and $P(\text{loss} > 3\%)$;
- * plot the point against the full distribution.

Match every printed number to the slides.

Check against the slides

- RMSE naive 1.04, AR 0.74.
- MAE naive 1.00, AR 0.65.
- 5% worst case = -2.79%.
- $P(\text{loss} > 3\%) = 4\%$.

Deliverable: `part1_practice.py` whose printed numbers match the hand calculations, then the charts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you decide which forecast is better?

**Lower out-of-sample
RMSE/MAE**

Q2 Why is the naive forecast important?

**It's the benchmark
every model must
beat**

Q3 Why can't a point forecast set risk capital?

**It says nothing about
the tail**

All three come from the held-out scoreboard, the naive bar, and the tail calculation.

Three Pitfalls to Avoid

Where model choice slips

Common mistakes

- "The best in-sample fit is the best model."
- "A fancier model must forecast better."
- "A point forecast is enough for any decision."

The reality

- In-sample fit flatters; grade out-of-sample.
- Beat the naive benchmark, or simplify.
- Risk decisions need the whole distribution.

Two Calculations to Remember

Part 1 applications in one screen

1

Grade forecasts out-of-sample

Naive and AR(1) scored RMSE 1.04 vs 0.74 on four held-out days. AR(1) won here - but the discipline is what matters: fit on the past, grade on the future, and beat the naive benchmark or simplify.

2

A risk decision needs the distribution

A point forecast of +0.5% is silent on risk. The distribution gives a 5% worst case of -2.79% and a 4% chance of losing more than 3% - the numbers a risk limit is actually built on.

3

Start from the decision

Match the forecast shape - point, regime, or distribution - to what the decision needs, then let out-of-sample testing settle any remaining choice.

Next — Part 2: the fair comparison, applying all three families to one asset and testing them.